

Triangles from Rectangles



Grade 6 • Mathematics
NC.6.G.1

Use decomposition to find triangle area.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

A triangular display has base 8 cm and perpendicular height 7 cm. Find the area. Explain why multiplying base by height alone is too large.

$A = \frac{1}{2} \times 8 \times 7 = 28 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.



Build a second model together

A triangular display has base 6 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.

Discuss: What should the first step find?



Your turn • 1

A triangular display has base 12 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.

Represent → solve → explain → check



Your turn • 2

A triangular display has base 14 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.

Represent → solve → explain → check



Your turn • 3

A triangular display has base 8 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.

Represent → solve → explain → check



Your turn • 4

A triangular display has base 8 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.

Represent → solve → explain → check



Exit question • Explain independently

A triangular display has base 12 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.

Include a check. What would you teach someone else?



Teacher review and next steps

1. 24 cm²

2. 42 cm²

3. 16 cm²

4. 8 cm²

Exit: 12 cm²

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

